

tf.keras.layers.UpSampling2D crashes(aborts) when size is large #46914

Closed

#51497

DNXie opened on Feb 4, 2021

System information

- Have I written custom code (as opposed to using a stock example script provided in TensorFlow): No
- OS Platform and Distribution (e.g., Linux Ubuntu 16.04): Linux Ubuntu 18.04
- Mobile device (e.g. iPhone 8, Pixel 2, Samsung Galaxy) if the issue happens on mobile device: N/A
- TensorFlow installed from (source or binary): binary
- TensorFlow version (use command below):2.1.0
- Python version:3.7.6
- Bazel version (if compiling from source):N/A
- GCC/Compiler version (if compiling from source):N/A
- CUDA/cuDNN version:N/A
- GPU model and memory:N/A

Describe the current behavior

`tf.keras.layers.UpSampling2D` crashes(aborts) when `size` is large

Describe the expected behavior

expect an exception message if the input unexpected instead of crash.

Standalone code to reproduce the issue

```
import tensorflow as tf
import numpy as np
tf.keras.layers.UpSampling2D(size=1610637938, data_format='channels_first', interpolation='b
```

Output:

```
2021-02-04 04:44:48.936606: F tensorflow/core/framework/tensor_shape.cc:353] Check failed: 0
Aborted (core dumped)
```



DNXie added type:bug on Feb 4, 2021

 **google-mi-butler** assigned amahendrakar on Feb 4, 2021

 amahendrakar on Feb 5, 2021

Contributor ...

Was able to reproduce the issue with TF v2.3, TF v2.4 and TF-nightly. Please find the gist of it [here](#). Thanks!

 **amahendrakar** added **comp:keras** **TF 2.4** on Feb 5, 2021

 **amahendrakar** assigned ymodak and unassigned amahendrakar on Feb 5, 2021

 **ymodak** added **stat:awaiting tens...** on Feb 5, 2021

 saikumarchalla on May 29, 2021

...

Colab crashes till in TF 2.6 Nightly as well. Please find the gist [here](#). Thanks!

 **yongtang** mentioned this on Aug 15, 2021

 [Fix crash with tf.image.resize if size is large #51497](#)

 yongtang on Aug 15, 2021

Member ...

Added a PR [#51497](#) for the fix.

 **copybara-service** closed this as completed in [#51497](#) on Aug 18, 2021

 google-mi-butler **bot** on Aug 18, 2021

...

Are you satisfied with the resolution of your issue?

[Yes](#)

[No](#)

 **rasabot** mentioned this on Nov 13, 2021

 [Security Alert RasaHQ/rasa#8638](#)

 oawxkw on Sep 12, 2023 · edited by oawxkw

Edits ...

I also observed the following API aliases or similar APIs can cause the same issue in older versions of tensorflow.

Users should be cautious when using them on both CPU and GPU up to tensorflow 2.6.0 (v2.6.0-rc2-32-g919f693420e).

- `(tf.keras.layers.UpSampling2D)`, `tf.compat.v1.keras.layers.UpSampling2D`
- `tf.keras.layers.UpSampling1D`, `tf.compat.v1.keras.layers.UpSampling1D`
- `tf.keras.layers.UpSampling3D`, `tf.compat.v1.keras.layers.UpSampling3D`

► Code to reproduce the issue for above APIs in older versions

It seems to be fixed in tensorflow 2.6.1 (v2.6.0-101-g3aa40c3ce9d) and later versions.

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Assignees



ymodak

Labels

TF 2.4

comp:keras

stat:awaiting tensorflower

type:bug

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development



Code with agent mode



Fix crash with `tf.image.resize` if size is large

tensorflow/tensorflow

Participants



+1

